Advanced Econometrics: Exercise #9

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Problem 1

Obtain the 95% confidence interval for the **mean** of lhhexp1 using:

i. Standard asymptotic theory

Solution

sum lhhexp1

```
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     lhhexp1 |      5,999    9.341561    .6877458   6.543108   12.20242
```

ci means lhhexp1

```
    Variable |        Obs        Mean    Std. err.       [95% conf. interval]
-------------+---------------------------------------------------------------
     lhhexp1 |      5,999    9.341561    .0088795        9.324154    9.358968
```

The 95% confidence interval is [9.324154, 9.358968]

ii. Standard asymptotic theory using the boostrap mean and the boostrap standard errors

Solution

bootstrap r(mean), reps(499): sum lhhexp1

First generate bootstrap sample with 499 repetitions at the mean then showing the summary.

```
Bootstrap results                                         Number of obs = 5,999
                                                          Replications  =   499

      Command: summarize lhhexp1
        _bs_1: r(mean)

------------------------------------------------------------------------------
             |   Observed   Bootstrap                         Normal-based
             | coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |   9.341561   .0089509  1043.65   0.000     9.324018    9.359105
------------------------------------------------------------------------------
```

The 95% confidence interval is [9.324018, 9.359105].

iii. The boostrap percentile method

Solution

estat bootstrap, all

This is asking STATA to show the information of all bootstrap confidence interval construction methods. The middle line is the bootstrap confidence interval using the percentile method.

Problem 1 continued on next page...

```
Bootstrap results                               Number of obs     =      5,999
                                                Replications      =        499

      Command: summarize lhhexp1
        _bs_1: r(mean)

------------------------------------------------------------------------------
             |   Observed               Bootstrap
             | coefficient       Bias    std. err.  [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |    9.3415612   .0001288   .00895086    9.324018   9.359105   (N)
             |                                        9.32274   9.359476   (P)
             |                                        9.323156   9.359813  (BC)
------------------------------------------------------------------------------
Key:  N: Normal
      P: Percentile
     BC: Bias-corrected
```

The 95% confidence interval is [9.32274, 9.359476]

iv. The boostrap t-statistics with both methods

Solution

Minimizing $S(b) = \frac{1}{N}\sum_{i=1}^{N}(Y - b)^2 \rightarrow$ taking the derivative and setting to zero

$$\frac{2}{N}\sum_{i=1}^{N}(Y_i - \hat{b}) = 0$$
$$\frac{2}{N}\sum_{i=1}^{N}Y_i - \frac{2}{N}\sum_{i=1}^{N}\hat{b} = 0$$
$$\frac{2}{N}\sum_{i=1}^{N}Y_i - \frac{2}{N}N\hat{b} = 0$$
$$2\hat{b} = 2\frac{1}{N}\sum_{i=1}^{N}Y_i$$
$$\hat{b} = \frac{1}{N}\sum_{i=1}^{N}Y_i$$

which is just the sample mean.

1. bootstrap "regress lhhexp1" _b _se, reps(499) saving(bsdata)

```
Command:      regress lhhexp1
Statistics:   b_cons      = _b[_cons]
              se_cons     = _se[_cons]

Bootstrap statistics                              Number of obs    =      5999
                                                  Replications     =       499

------------------------------------------------------------------------------
Variable     |  Reps  Observed      Bias  Std. err. [95% conf. interval]
-------------+----------------------------------------------------------------
      b_cons |   499  9.341561  .0001287  .0089509   9.323975   9.359147   (N)
             |                                        9.32274   9.359476   (P)
             |                                       9.323156   9.359813  (BC)
     se_cons |   499  .0088795  3.52e-06  .0000952   .0086924   .0090666   (N)
             |                                        .008699   .0090717   (P)
             |                                       .0086904   .0090685  (BC)
------------------------------------------------------------------------------
Key:  N: Normal
      P: Percentile
     BC: Bias-corrected
```

2. regress lhhexp1

```
      Source |       SS           df       MS      Number of obs   =     5,999
-------------+----------------------------------   F(0, 5998)      =      0.00
       Model |           0         0           .   Prob > F        =         .
    Residual |  2837.01998     5,998  .472994328   R-squared       =    0.0000
-------------+----------------------------------   Adj R-squared   =    0.0000
       Total |  2837.01998     5,998  .472994328   Root MSE        =    .68775

------------------------------------------------------------------------------
     lhhexp1 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _cons |   9.341561   .0088795  1052.04   0.000     9.324154    9.358968
------------------------------------------------------------------------------
```

3. Now use the saved boostrap dataset to continue, in this case, bsdata.dta

The first method is the **non-symmetric test** with the critical values of $cv_{boot}(\frac{\alpha}{2})$ and $cv_{boot}(1-\frac{\alpha}{2})$ for $(\hat{t}_1, \ldots, \hat{t}_B)$.

1. gen double t_cons = (b_cons-_b[_cons])/ se_cons
2. _pctile t_cons, percentiles(2.5 97.5)
3. display _b[_cons]-_se[_cons]*r(r2)
4. display _b[_cons]-_se[_cons]*r(r1)

The result gives the 95% confidence interval of [9.3238664, 9.3603002].

The second method is the **symmetric test** with the critical value of $cv^a_{boot}(1-\alpha)$ for $(|\hat{t}_1|, \ldots, |\hat{t}_B|)$.

1. gen double at_cons=abs(t_cons)
2. _pctile at_cons, percentiles(95)
3. display _b[_cons]-_se[_cons]*r(r1)
4. display _b[_cons]+_se[_cons]*r(r1)

The result gives the 95% confidence interval of [9.3229329, 9.3601895].

Problem 2

Suppose $X_1, \ldots, X_n$ are i.i.d. continuous r.v. from distribution with cumulative distribution function $F_X(\cdot)$ and probability density function $f_X(\cdot)$. Let M_n be the **sample median** and m_0 be the **population median**. In this case $\sqrt{n}(M_n - m_0) \xrightarrow{d} N\left(0, \frac{1}{4(f_X(m_0))^2}\right)$.

i. Propose a consistent estimator for the asymptotic variance of $\sqrt{n}(M_n - m_0)$

Solution

A consistent estimator for the density is $\frac{1}{4\hat{f}_x^2(M_N)}$ where $\hat{f}_x(M_N) = \frac{1}{Nh}\sum_{i=1}^{N} K\left(\frac{y_i - M_N}{h}\right)$, a kernal density estimator.

ii. Obtain a 95% CI for the median of lhhexp1

A. using standard asymptotic theory with the consistent estimator proposed previously

Solution

$SE(M_N) = \frac{1}{\sqrt{n}\sqrt{4f_X^2(M_N)}}$

$(M_N - Z_{\frac{\alpha}{2}}\, SE(M_N),\, M_N + Z_{\frac{\alpha}{2}}\, SE(M_N))$

$P(Z > Z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$, $Z \sim N(0,1)$

$h = 0.09904 \rightarrow$ obtained from Exercise sheet 8 with **Silverman's rule of thumb**

Steps to do:

1. Find the median
2. Generate the sequence
3. Find the mean of sequence

1. Do not forget to clear the dataset first as in to reload the original dataset
2. sum lhhexp1, d

```
                            lhhexp1
-------------------------------------------------------------
      Percentiles      Smallest
 1%     7.586134       6.543108
 5%     8.254869       6.818342
10%     8.507335       6.892809       Obs               5,999
25%     8.919547       6.965246       Sum of wgt.       5,999

50%     9.311416                      Mean           9.341561
                        Largest       Std. dev.      .6877458
75%     9.759566       11.73528
90%     10.22297       11.75362       Variance       .4729943
95%     10.51927       12.03334       Skewness        .109243
99%     11.11577       12.20242       Kurtosis       3.599002

. * Median = 9.311416
```

3. gen double a = normalden((lhhexp1 -9.311416)/0.09904)/0.09904
4. mean a

```
Mean estimation                          Number of obs = 5,999

--------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+------------------------------------------------
           a |   .6392072   .0153739      .6090688    .6693457
--------------------------------------------------------------
```

5. display (1/(2*.6392072*sqrt(5999))) to get $SE(M_N)$
6. display 9.311416-1.96*0.01009925 to get lower bound of CI
7. display 9.311416+1.96*0.01009925 to get upper bound of CI

The 95% confidence interval is [9.2916215, 9.3312105]

B. using standard asymptotic theory with the bootstrap mean and bootstrap standard error

Solution

bootstrap r(p50), reps(499): sum lhhexp1, d

```
Bootstrap results                                   Number of obs = 5,999
                                                    Replications  =   499

      Command: summarize lhhexp1, d
        _bs_1: r(p50)

------------------------------------------------------------------------------
             |   Observed   Bootstrap                         Normal-based
             | coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |   9.311416   .0113762   818.50   0.000     9.289119    9.333713
------------------------------------------------------------------------------
```

C. using the boostrap percentile method

Solution

estat bootstrap, all

```
Bootstrap results                               Number of obs     =      5,999
                                                Replications      =        499

      Command: summarize lhhexp1, d
        _bs_1: r(p50)

------------------------------------------------------------------------------
             |    Observed               Bootstrap
             | coefficient       Bias    std. err.  [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |   9.3114157  -.0005968   .01137624    9.289119   9.333713   (N)
             |                                       9.286874   9.333403   (P)
             |                                       9.289034   9.334837  (BC)
------------------------------------------------------------------------------
Key:  N: Normal
      P: Percentile
     BC: Bias-corrected
```